

BENCHMARK: INACCURATE

Results

Mid-Transit Time (Tmid)	2460344.7979 +/- 0.0011 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.1326 +/- 0.0097
Transit depth (Rp/Rs)^2	1.76 +/- 0.26 [%]
Orbital Inclination (inc)	86.78 +/- 2.82
Airmass coefficient 1 (a1)	1.028 +/- 0.011
Airmass coefficient 2 (a2)	-0.0204 +/- 0.0039
Scatter in the residuals of the lightcurve fit is	1.11 %
Best Comparison Star	#3 - [305, 265]
Optimal Aperture	3.7
Optimal Annulus	13.5
Transit Duration (day)	0.1302 +/- 0.003

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R fit vs published	0.1326 ± 0.0097 vs 0.1178 (deviation 1.51σ)
Duration fit vs published	0.1302 d vs 0.125 d (deviation 1.72σ)
Mid-time O-C	1573.6 min
Depth SNR	13.5
Residual scatter	1.11 %

Light curve

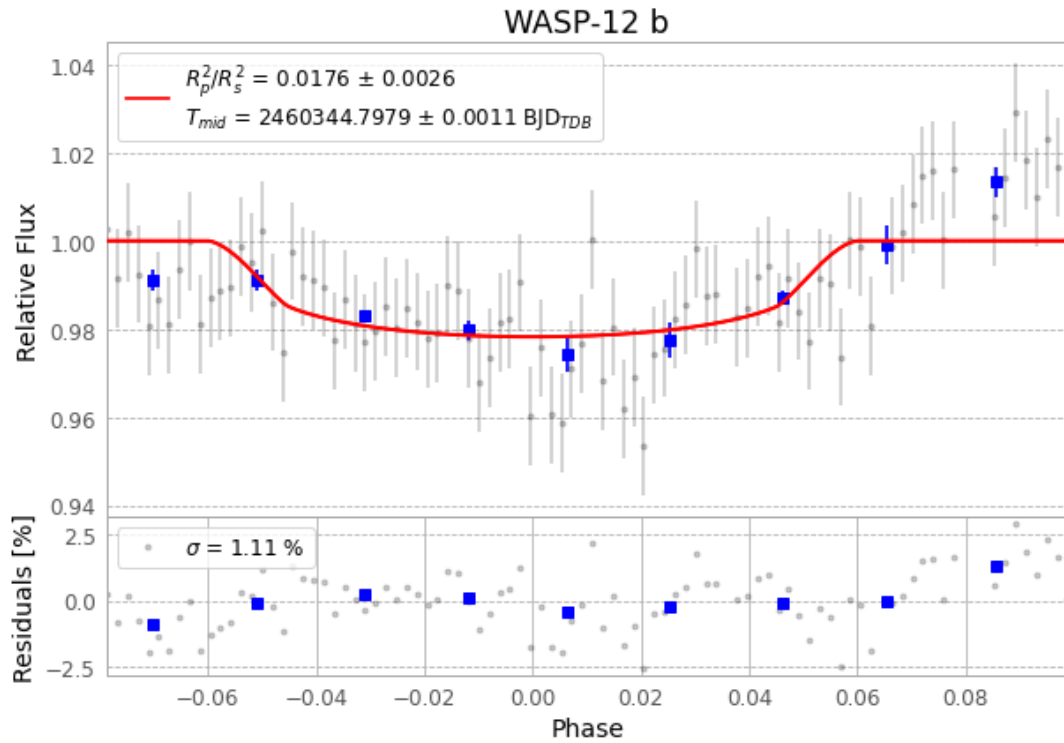


Figure 1 — FinalLightCurve_WASP-12 b_2024-02-03.png (SHA-256 a98f1b96c5416f2d...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [384, 227], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 8f45170761a01987...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit 8bd826b

Data provenance

94 FITS frames (62 MB), WASP-12240204045715.FITS → WASP-12240204093615.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-WASP-12-240203.log (a6ef6da06ad5...), FinalLightCurve_WASP-12 b_2024-02-03.png (a98f1b96c541...), FinalLightCurve_WASP-12 b_2024-02-03.csv (3fc614cdf228...)

Compute resources

Wall time 150.1 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-07-17 22:45Z.